

RSnowflake in Snowflake Workspace Notebooks

This vignette focuses on **Snowflake Workspace Notebooks**: how RSnowflake fits the Python + %%R model, what to install first, and how **ADBC** (`adbcsnowflake`) behaves behind the internal SPCS gateway. For general DBI usage and authentication modes, see `vignette("getting-started", package = "RSnowflake")`.

Execution model

1. The **Python kernel** holds the Snowpark session Workspace creates for you.
2. **%%R cells** run R via rpy2; RSnowflake in R uses HTTPS like anywhere else.
3. `dbConnect(Snowflake())` in R succeeds when the environment variables `Workspace / setup_notebook()` export are present (see the getting-started vignette, **Snowflake Workspace Notebooks** section).

Run your **Python bootstrap** (for example `setup_notebook()` from `sfnb_setup.py`) **before** the first **%%R** cell that loads RSnowflake.

Optional ADBC (adbcsnowflake)

ADBC is **optional**. When `adbcsnowflake` and `adbcdrivermanager` are installed, RSnowflake can use Arrow-native paths for reads and bulk writes. Building `adbcsnowflake` requires a **Go compiler** on the machine where `install.packages()` runs.

In **Workspace**, install tooling from the **Python / micromamba** side so `PATH` and compilers are consistent. Do **not** rely on a minimal `PATH` inside a bare **%%R** cell to find Go: follow the bootstrap script for your stack (many teams use `sfnb-multilang` with an `--adbc` or equivalent preset).

External Access Integration (EAI) must allow every HTTP redirect needed to download R packages and Go modules (for example R-universe / CDN and `proxy.golang.org / storage.googleapis.com`). Your Snowflake admin can compare hosts against network-rule documentation for the snowflake-notebook-multilang toolkit.

Internal SPCS gateway (SNOWFLAKE_HOST)

From Workspace containers, RSnowflake and ADBC can use the **internal** gateway (`SNOWFLAKE_HOST`) with the container **OAuth** token. You do **not** need the public `*.snowflakecomputing.com` URL for routine Snowflake traffic when the session is wired correctly.

Bulk write routing (`upload_method = "auto"`, thresholds, fallbacks) is described in `vignette("getting-started", package = "RSnowflake")` (**Bulk Write Methods**). Tuning options use R options such as `RSnowflake.upload_method` and `RSnowflake.bulk_write_threshold`.

Deep dive and benchmarks

The authoritative engineering note for Workspace + ADBC (architecture, redirect chains, benchmarks, PAT fallbacks for **external** hosts) ships in the RSnowflake repository as `WORKSPACE_ADBC.md`:

https://github.com/Snowflake-Labs/RSnowflake/blob/main/WORKSPACE_ADBC.md

Companion ML workflows

If you use `snowflakeR` in the same notebook, DBI access from R can share credentials via `snowflakeR::sfr_db_connection()`. See `vignette("workspace-notebooks", package = "snowflakeR")`.